

Introduction to Pandas with SQLite

```
In [ ]: import pandas as pd
import sqlite3
import warnings
warnings.filterwarnings("ignore")
```

```
/tmp/ipykernel_17605/501431013.py:1: DeprecationWarning:
Pyarrow will become a required dependency of pandas in the next major release of pandas (pandas 3.0),
(to allow more performant data types, such as the Arrow string type, and better interoperability with other libraries)
but was not found to be installed on your system.
If this would cause problems for you,
please provide us feedback at https://github.com/pandas-dev/pandas/issues/54466

import pandas as pd
```

Create a connection

```
In [ ]: conn = sqlite3.connect("Example3.db")
curr = conn.cursor()
curr.execute("SELECT * FROM Customers")
curr.fetchall()
```

```
Out[ ]: [('Vishwas', 'Singh', 30, 35000.0),
('Shri', 'Tripathi', 30, 3000.0),
('sai', 'singh', 27, 4000.0),
('Komal', 'singh', 25, 50000.0)]
```

Creation of a DataFrame

```
In [ ]: df = pd.read_sql_query("SELECT * FROM Customers", conn)
```

```
In [ ]: df
```

```
Out[ ]:
```

	FirstName	LastName	Age	Salary
0	Vishwas	Singh	30	35000.0
1	Shri	Tripathi	30	3000.0
2	sai	singh	27	4000.0
3	Komal	singh	25	50000.0

Run Queries from pandas

```
In [ ]: pd.read_sql("SELECT * FROM customers",conn)
```

```
Out[ ]:
```

	FirstName	LastName	Age	Salary
0	Vishwas	Singh	30	35000.0
1	Shri	Tripathi	30	3000.0
2	sai	singh	27	4000.0
3	Komal	singh	25	50000.0

```
In [ ]: pd.read_sql("SELECT * FROM customers WHERE LastName='Singh'",conn)
```

```
Out[ ]:
```

	FirstName	LastName	Age	Salary
0	Vishwas	Singh	30	35000.0

```
In [ ]: pd.read_sql("SELECT FirstName, LastName FROM customers",conn)
```

```
Out[ ]:
```

	FirstName	LastName
0	Vishwas	Singh
1	Shri	Tripathi
2	sai	singh
3	Komal	singh

Read CSV From Pandas

```
In [ ]: sample_df = pd.read_csv("sample.csv")
```

```
In [ ]: type(sample_df)
```

```
Out[ ]: pandas.core.frame.DataFrame
```

```
In [ ]: sample_df.head(2)
```

```
Out[ ]:
```

	Name	Age
0	Vishwas	30
1	Swetha	25

```
In [ ]: sample_df.tail(2)
```

```
Out[ ]:
```

	Name	Age
4	Sandeep	33
5	Rohith	33

Creating a table in SQLite from a CSV File with Pandas

```
In [ ]: sample_df = pd.read_csv("sample.csv")
```

```
In [ ]: sample_df.columns
```

```
Out[ ]: Index(['Name', 'Age'], dtype='object')
```

```
In [ ]: sample_df.to_sql("sample_table",conn,if_exists="replace",index=False)
```

```
Out[ ]: 6
```

```
In [ ]: pd.read_sql_query("SELECT * FROM sample_table",conn)
```

```
Out[ ]:
```

	Name	Age
0	Vishwas	30
1	Swetha	25
2	Sri	30
3	Ana	26
4	Sandeep	33
5	Rohith	33

Function to get all the tables in a database

```
In [ ]: conn = sqlite3.connect("Example3.db")  
curr = conn.cursor()
```

```
In [ ]: def table_info(conn,cursor):
        '''Prints out all of the columns
           of every table in DB

           conn: database connection object
           cursor: Cursor Object
        '''

        tables = cursor.execute("SELECT name FROM sqlite_master WHERE type='table';").fetchall()
        for table_name in tables:
            table_name = table_name[0]
            table = pd.read_sql_query("SELECT * FROM {} LIMIT 0".format(table_name),conn)
            print(table_name)
            for col in table.columns:
                print("\t "+col)
            print()
```

```
In [ ]: table_info(conn,curr)
```

```
customers
    FirstName
    LastName
    Age
    Salary

sample_table
    Name
    Age
```

See Table schema with SQLite

```
In [ ]: curr.execute("PRAGMA table_info('sample_table')").fetchall()
```

```
Out[ ]: [(0, 'Name', 'TEXT', 0, None, 0), (1, 'Age', 'INTEGER', 0, None, 0)]
```

```
In [ ]:
```